

**CPT6314 Final Year Project (FYP) 1 Meeting Log
Trimester OCT / NOV 2025 (Trimester ID:2530)**

Meeting Date: 21-1-2025	Meeting No.: 5
Meeting Mode: Face-to-Face	
Project ID: 860	Project Type: Research-based
Project Title : Autonomous Driving Agent Training Using Deep Reinforcement Learning in CARLA Simulator	
Student ID : 242UC243LS	Student Name: NG JUN XUAN
Student Programme and Specialisation: Data Science	
Supervisor Name: Dr. Mohammad Shadab Khan	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

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1. WORK DONE

Tasks: Background Study or Literature Review / Design or Research Methodology / Proof of Concept

Details (in point form):

- 4.1. Introduction
- 4.2. Comparative Analysis of Merging and Intersection Scenarios
- 4.3. Comparative Analysis of Roundabout and Intersection
- 4.4. Comparative Analysis of Intersection and Highway
- 4.5. Comparative Analysis of Merging, Roundabout and Intersection
- 4.6. Comparative Analysis of Merging, Intersection and Highway
- 4.7. Analysis summary
- 5.1. Introduction
- 5.2. Vision Transformer
- 5.3. Decision Transformer
- 5.4. General Transformer
- 5.5. Other Transformers

2. WORK TO BE DONE

Tasks: Draft Report Completion / Problem Formulation and Project Planning / Design or Research Methodology

Details (in point form):

- Chapter 1: Introduction
- Chapter 6: Implementation Plan
 - Details of the equipment to be used
 - Description of the algorithms, scenarios, and security metrics to be used in FYP2
- Chapter 7: Conclusion and Outlook
 - Answers to all research questions
 - Outlook for FYP2

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

Problem:

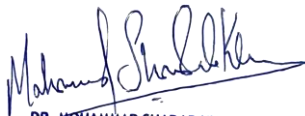
- Establishing a standardized method to quantify the risk levels of diverse driving scenarios (e.g., Roundabouts, Merging, Intersections). There is significant variability in the risk assessment metrics used across different primary studies.
- Systematically categorizing and summarizing the wide array of Transformer variants, especially hybrid architectures that integrate CNNs with self-attention mechanisms.

Solution:

- Consistent with the algorithmic comparison methodology, scenario risk is evaluated by comparing the Collision Rate or Success Rate within the same study across different environments. By extracting the relative difficulty levels from these specific contexts, a unified scenario risk hierarchy can be synthesized.
- Models are classified based on their core input modalities and functions: those processing image patches are categorized as Vision Transformer (ViT), while those modeling sequential decisions are grouped under Decision Transformer (DT). Any architecture lacking a specific variant designation or employing standard self-attention mechanisms is consolidated under the term "General Transformer" to ensure analytical clarity.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

Satisfactory Progress.



DR. MOHAMMAD SHADAB KHAN
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Supervisor's Signature



Student's Signature

Co-Supervisor's Signature
(if applicable)

Company Supervisor's Signature
(if applicable)

IMPORTANT NOTES TO STUDENTS:

1. Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
2. Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
Minimum requirement is SIX Meeting Logs (Period: Week 3 to Week 12). Students can have fortnightly meetings with the supervisor.
3. Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.

This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.

4. Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.